

Ajay I

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Summary

Computer Science Engineering student skilled in predictive machine learning and building AI agents. Proficient in Python, Java, C and C++, with a proven track record in competitive hackathons and a research publication. Passionate about leveraging advanced ML models and robust architectures to develop scalable, data-driven solutions for real-world challenges.

Education

Sri Ramakrishna Engineering College, Coimbatore

2024 – 2028

B.E. Computer Science and Engineering (Anna University) | CGPA: 9.08 / 10 (up to 4th Sem)

Coimbatore, IN

Githanjali Public School, Coimbatore (CBSE)

2022 – 2024

HSC (2024) | SSLC (2022)

Technical Skills

→ **Languages:** Java, C, C++, Python

→ **AI / ML & Data:** Predictive Machine Learning, NLP (Transformers), Multi-Agent AI, CatBoost, XGBoost, Isolation Forest, Pandas, NumPy, NetworkX, Data Analytics & Visualization, Geospatial Processing

→ **Web & Backend:** FastAPI, React.js, Tailwind CSS, JavaScript, REST APIs, GitHub / Slack / Twilio / WhatsApp APIs

→ **Databases & Core:** Relational DBs, SQLite, PostgreSQL, Firebase | Problem Solving, Full-Stack Development, Rapid Prototyping

Projects

Workforce Contribution Monitor

January 2026

→ Built a full-stack evaluation system that integrates communication (**Slack**) and execution (**GitHub**) platform data through automated API-based **ingestion pipelines** to bridge the gap between employee activity volume and actual work impact, using a modular backend with efficient database handling for scalability and real-time processing.

→ Designed a **custom scoring algorithm** that separates high-activity from high-impact contributions and presented the results on an interactive, responsive dashboard of contribution patterns, trends and performance indicators.

Tech Stack: React, Tailwind CSS, Python, FastAPI, SQLite, GitHub & Slack APIs

AI Customer Feedback Intelligence & Retention System

→ Developed a system that analyzes multi-channel customer feedback using **Transformer-based NLP** sentiment analysis and behavioral modeling to detect dissatisfaction, identify at-risk users and predict churn.

→ Automated a personalized retention pipeline that triggers email and WhatsApp follow-ups by risk level, and integrated a **multi-agent decision framework** that recommends refunds, escalations or retention offers to improve decision efficiency.

Tech Stack: Python, NLP (Transformers), FastAPI, React, Twilio API / WhatsApp API, PostgreSQL

Satellite-Based Water Quality Monitoring & Prediction System (Mini Project)

→ Developed a system that processes satellite imagery and geospatial data to predict key water quality parameters with a **CatBoost Regressor**, using engineered spectral and environmental features, enabling large-scale monitoring without manual sampling.

→ Built an automated ingestion, preprocessing and prediction pipeline with visualization modules for spatial and temporal trend analysis.

Tech Stack: Python, CatBoost, Pandas, NumPy, Geospatial Processing, Data Visualization Tools

ChainWatch – Blockchain AML Intelligence System

September 2026

→ Developed a real-time system to detect suspicious cryptocurrency transactions using **XGBoost**, **Isolation Forest**, graph analytics and multi-agent pattern detection, identifying high-risk wallet activity and generating explainable risk scores.

Tech Stack: Python, FastAPI, XGBoost, Isolation Forest, NetworkX, Firebase, JavaScript

Publications

[A High-Resolution Data Fusion Framework for Water Body Assessment Using Sentinel-2 and Gradient Boosting](#)

[IEEE Xplore](#) | [DOI](#) | ICCPCT 2026 | IEEE Kerala | 2026

Hackathons & Achievements

→ **First Prize** — 24-hr HackTIDE Hackathon, Amrita Vishwa Vidyapeetham, Coimbatore (08.01.2026) | ₹25,000

→ **Second Prize** — 12-hr Hackathon, Sri Ramakrishna College of Arts & Science (SRCAS), Coimbatore (17.02.2025) | ₹2,500

→ **Third Prize** — 24-hr VeloHack'26, Vel Tech University, Chennai (11.09.2026) | ₹16,000

Certifications

→ **Introduction to Artificial Intelligence** — Infosys Springboard

→ Cleared **Business English Certificate (BEC)** — Business Preliminary Examination